ImmunoBlitz: Game vs. Real Biology

Scientific Model Evaluation Activity - compare ImmunoBlitz game mechanics with the real biology they represent.

Name: _____ Class: _____ Date: _____

Key idea: ImmunoBlitz is an educational model. A useful model does not need to copy every detail of reality, but we should be able to explain what it represents, what it simplifies, and where a player could misunderstand the science.

Part A - Choose Three Game Mechanics

Select three things you noticed while playing. Examples include pathogen bubbles, waves, organ health, skin or mucus upgrades, white blood cells, vaccines, antibiotics, masks, antiseptics, or another mechanic.

Game mechanic	What real biological idea does it represent?	What has the game simplified or exaggerated?
1.		
2.		
3.		

Part B - Look More Closely at One Mechanic

Choose the most interesting mechanic from Part A. Now evaluate it like a scientist evaluates a model.

1. The mechanic I chose:

Name the mechanic clearly.

2. What does the game get right?

Describe one idea that the mechanic communicates accurately or usefully.

Part B - Evaluate the Model

3. What does the game simplify?

Explain what is missing, compressed, instant, exaggerated, or made easier to understand than in real biology.

4. Could a player misunderstand anything?

Describe one incorrect idea someone might develop if they treated the game mechanic as a literal picture of real biology.

5. How would you improve the mechanic?

Suggest one change that would make the science clearer without making the game too complicated or less enjoyable.

Part C - Apply Your Science Knowledge

1. Which defences happen automatically, and which require a human decision?

Use examples from the game such as barriers, immune cells, vaccination, medicines, hygiene, masks or other interventions.

2. Why can a simplified model still be useful in science?

Explain why scientists and teachers use models even when they leave out details.

Part D - Final Judgement

Complete this statement: "ImmunoBlitz is a useful model for learning about infection and immunity because... However, it should not be treated as a literal simulation because..."

Remember: Scientists use models because reality is often too complex or difficult to observe directly. Good scientific thinking includes knowing both what a model helps us understand and where its limits are.